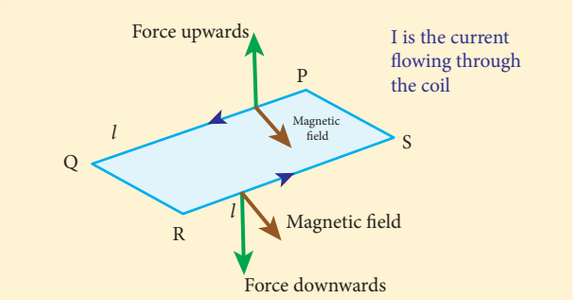


Figure 3.60 Force acting on current carrying coil

$$\tau = NABI \qquad (3.69)$$

Due to this deflecting torque, the coil gets twisted and restoring torque (also known as restoring couple) is developed. Hence the moment of the restoring torque is proportional to the amount of twist θ.Thus

$$\tau = K\theta \qquad (3.70)$$

where K is the restoring couple per unit twist.

At equilibrium, the deflecting couple must be equal to the restoring couple. Therefore we get,

$$NABI = K\,\theta$$

$$\Rightarrow I = \frac{K}{NAB}\theta \qquad (3.71)$$

(or) $I = G\,\theta$

where $G = \frac{K}{NAB}$ is called galvanometer constant or current reduction factor of the galvanometer.

Since the suspended moving coil galvanometer is very sensitive, we have to handle with high care while doing experiments. Most of the galvanometer we use are pointer type moving coil galvanometer.

Figure of merit of a galvanometer

It is defined as the current required to produce a deflection of one scale division in the galvanometer.

Sensitivity of a galvanometer

The galvanometer is said to be sensitive if it shows large scale deflection even for a small current passed through it or a small voltage applied across it.

Current sensitivity

It is defined as **the deflection produced per unit current flowing through galvanometer**.

$$I_S = \frac{\theta}{I} = \frac{NAB}{K} \Rightarrow I_S = \frac{1}{G} \qquad (3.72)$$

The current sensitivity of a galvanometer can be increased by

(i) increasing the number of turns, N
(ii) increasing the magnetic induction, B
(iii) increasing the area of the coil, A
(iv) decreasing the couple per unit twist of the suspension wire, K.

Phosphor - bronze wire is used as the suspension wire because the couple per unit twist is very small.

Voltage sensitivity

It is defined as **the deflection produced per unit voltage applied across galvanometer**.

$$V_S = \frac{\theta}{V}$$

$$V_S = \frac{\theta}{IR_g} = \frac{NAB}{KR_g} \quad \text{(or)}$$

$$V_S = \frac{1}{GR_g} = \frac{I_S}{R_g} \qquad (3.73)$$

where R_g is the resistance of galvanometer.

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